Experience Report

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Project: Marburg Cooking
Task assigned: Change-password feature on the user profile page

1. General

Motivation. I joined this module because I wanted to work on a real project together with people from other fields. Most of my courses are about theory or small assignments that one person can finish alone, and I wanted to see how a proper application gets built when several people with different backgrounds are involved. The Marburg Cooking project fit that well. It is a complete web application for cooking events in Marburg, with a backend, a frontend and a database, and it is meant to be used by real people. That made it feel more serious than a normal course exercise, and it was the main reason I picked it.

My role. My background is in computer science, so I worked on the technical side of the project. The task assigned to me was the change-password feature on the profile page. It is a small piece of the application, but it touches the frontend, the backend and the login flow, so it was a good way for me to understand how these parts depend on each other. Besides that, I looked at a duplicate call to the user endpoint that happened every time someone logged in, and I helped to remove it. My teammates come from other directions, so I tried to explain technical things in a simple way when they asked.

What I learned personally. The biggest lesson was how much work sits behind a feature that looks simple. A change-password form has two input fields and a button, but it also needs a request to the server, a check on the server side, a clear message for the user, and a login that behaves correctly afterwards. I also learned that testing a feature the way a normal user would use it finds problems that reading the code never shows. On the personal side, I learned that I have to ask earlier when I am unsure about something, instead of trying to solve everything alone.

2. Project course and organization

What I did. I implemented the option for a logged-in user to change their password. On the profile page there is now a "Change Password" section where the user types a new password and confirms it by typing it a second time. When the user clicks the update button, the React frontend sends a request to the Spring Boot backend. If the request is successful, the user is taken back to the "My Profile" page and sees the message "Password updated successfully". If something goes wrong, the user gets an error message instead, so nobody is left guessing what happened.

How I tested it. I ran the whole application on my own computer (backend, frontend and MySQL database) and went through the feature step by step like a normal user would. First I logged in, then I opened the profile page, changed the password, and logged out. After that, I tried to log in again with the old password. This was rejected with an error message, which is exactly what should happen. Then I logged in with the new password, and it worked and led me to the profile page. During the whole test I kept the browser's network tab open to check that the request to the backend returned a successful response. I recorded a short video of this test, so the group can see that the feature works from start to finish.

A problem I found on the way. While looking at the network tab, I noticed that the user data was requested twice after a login. It did not break anything, but it was unnecessary and made the app slower than needed. I traced it back to the login flow and fixed it, so the data is now requested only once.

Helping factors. Having a clear split of tasks helped a lot, because I could work on my part without waiting for anyone else. The shared setup also made things easier. We all used the same stack (Spring Boot, React with Vite and MySQL), so I could run and test my feature locally before it was combined with the rest of the project. Working in a group also meant that I could ask the others when something in the existing code was unclear, which saved me a lot of time compared to figuring it out alone.

Hindering factors. The hardest part for me was coordination. We come from different study programs and have different schedules, so it was difficult to know what everyone was working on at a given

moment. Some decisions were made in short conversations without being written down, and this made it harder to follow the overall progress. On the technical side, getting the backend to run locally took some effort because it needs specific startup settings, and small differences between our computers caused problems that were hard to find at first. I think a few fixed check-ins and a shared task list would have avoided most of this. Next time I would also push my work to the shared repository earlier and in smaller steps, so that the others can see and review it sooner.

Framework of the module. I think the idea of the module is good. The discussion rooms are a useful place to ask questions and to plan the project together with the instructors, and the freedom to choose a project was motivating. What made learning harder was mostly organization. It was not always clear which level of technical progress was expected at which point of the semester, and the connection between the module content and our actual work could be stronger.

Structure, content and supervision. The structure of the module (project work, presentations, reports) is clear on paper. The input sessions gave a useful general view of innovation and the regional ecosystem, but for a technical project I would have liked more links between the sessions and the work we do in the groups. The supervision was available when I needed it, but with so many groups it is hard to get detailed feedback on technical questions. For improvement, I would suggest short milestone meetings for each group and a simple guideline for how students from different disciplines can divide the work fairly.

3. Learning and skills gained

From this project I take away a much better understanding of how authentication works in a real application and how to test a feature properly before calling it finished. I also practiced working with an existing codebase that other people wrote, which is very different from starting a project alone. Reading other people's code, asking questions and fitting my work into a bigger system are skills that I will need in my master's studies and later in a job, where most work happens in teams and on code that already exists. I also became more aware of how important it is to communicate progress clearly, even when a task is small.

For the next steps of the project, I would suggest the following:

- **Stronger password rules:** a minimum length and some complexity requirements, shown directly in the form so that users know what is expected.
- **Ask for the current password:** a user should confirm the current password before a change is allowed, which protects the account if someone else uses the same device.
- **Automated tests:** tests for the change-password flow, so that later changes in the login code do not break it without anybody noticing.
- **Forgot-password option:** connect the feature with the "forgot password" link that is already on the login page, so both parts work together.